

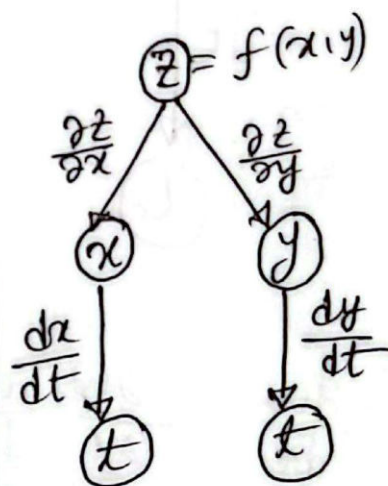
Lecture-09

* Chain rule for Partial Derivatives:
(H.A - section $\rightarrow$ 13.5.1)

- If $x = x(t)$ and $y = y(t)$ are differentiable at t , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t and,

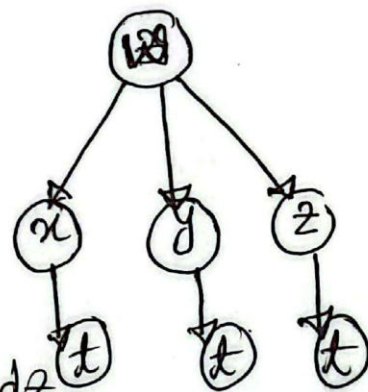
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y) .

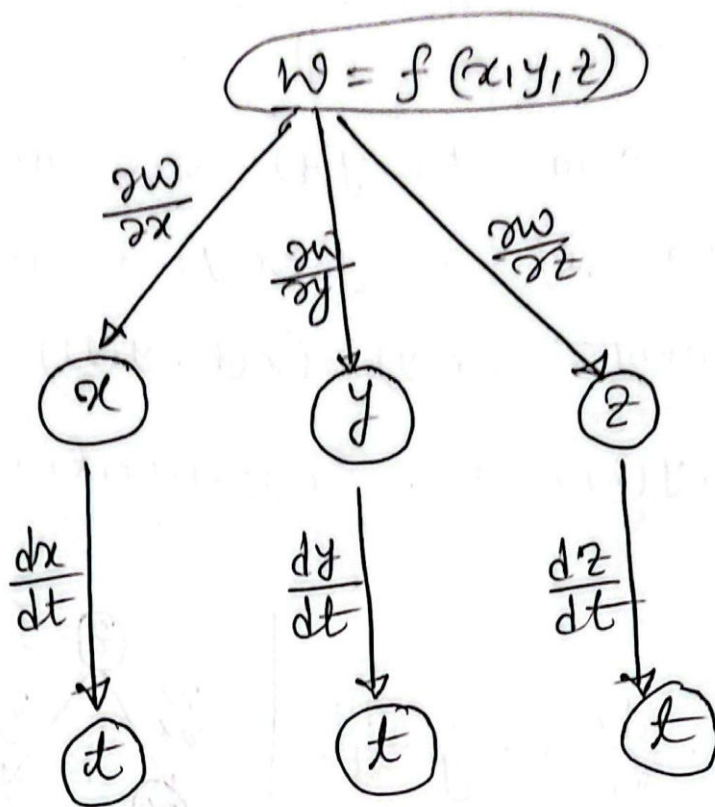


- For three independent variable function:
let, function is $w = f(x, y, z)$
and $x = x(t)$; $y = y(t)$;
 $z = z(t)$ are differentiable at t , then

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$



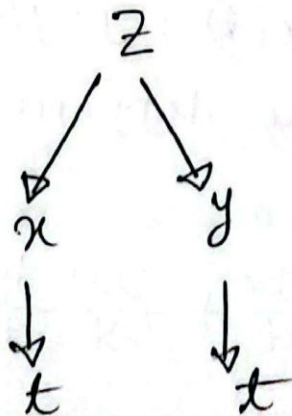
* Problem (03)



$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

* Example: If $z = 3x^2y^3$; $x = t^4$ and $y = t^2$ then evaluate $\frac{dz}{dt}$.

Ans:



$$\text{so, } \frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$= 6xy^3 \times 4t^3 + 9x^2y^2 \times 2t$$

$$= 6 \cdot t^4 \cdot (t^2)^3 \cdot 4t^3 + 9 (t^4)^2 (t^2)^2 \cdot 2t$$

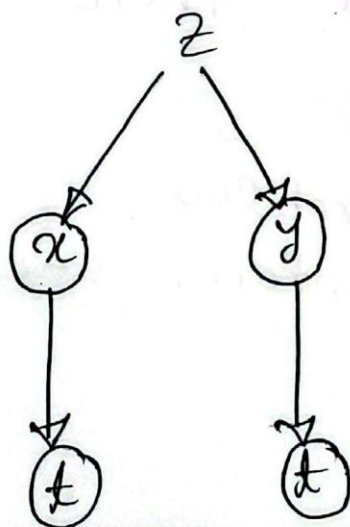
$$= 24 t^{13} + 18 t^{13}$$

$$\frac{dz}{dt} = 42 t^{13}$$

* Example: If $z = 3 \cos x - \sin xy$ and $x = \frac{1}{t}$, $y = 3t$, then evaluate $\frac{dz}{dt}$.

Ans:

Here z is a function of two independent variables and x and y are the functions of one independent variable t .



$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$= (-3 \sin x - y \cos xy) \left(\frac{-1}{t^2} \right) + (-x \cos xy) 3$$

$$= \frac{3 \sin\left(\frac{1}{t}\right) - 3t \cos\left(\frac{1}{t} \cdot 3t\right)}{t^2} + 3 \frac{1}{t} \cos\left(\frac{1}{t} \cdot 3t\right)$$

$$= \frac{3 \sin\left(\frac{1}{t}\right) - 3t \cos(3) + 3t \cos(3)}{t^2}$$

$$\frac{dz}{dt} = \frac{3 \sin(1/t)}{t^2}$$

* problem-(03)

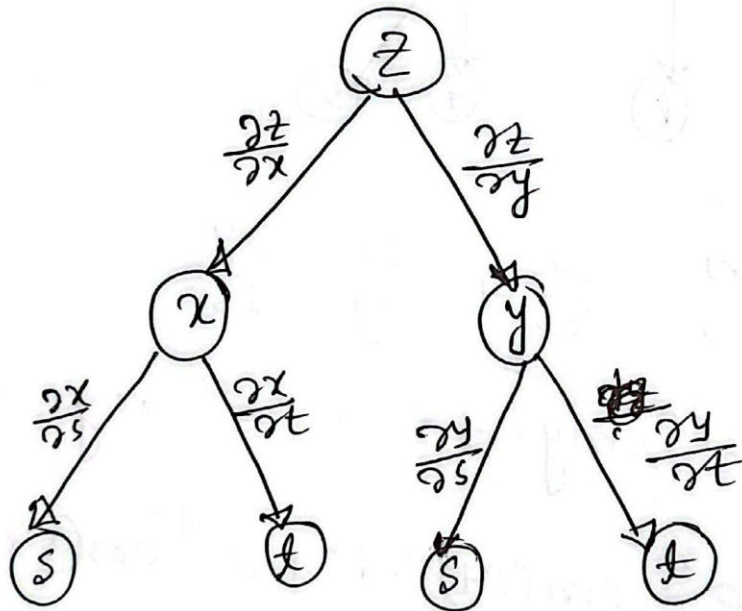
(i) If $z = xy e^y$ and $x = t^2$,
 $y = 5t$ then evaluate $\frac{dz}{dt}$

(ii) If $z = x^2 y + xy^2$ and $x = 3t$,
 $y = t^2$ then evaluate $\frac{dz}{dt}$.

* practice problem

H. Anton, section: 13.5 → 8 (1-10)

* the chain rule: (see theorem 13.5.2)
 let us suppose that $z = f(x, y)$ is
 a differentiable function of x and
 y where $x = x(s, t)$ and $y = y(s, t)$
 are differentiable functions of
 s and t . then,



$$\text{so, } \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t} ;$$

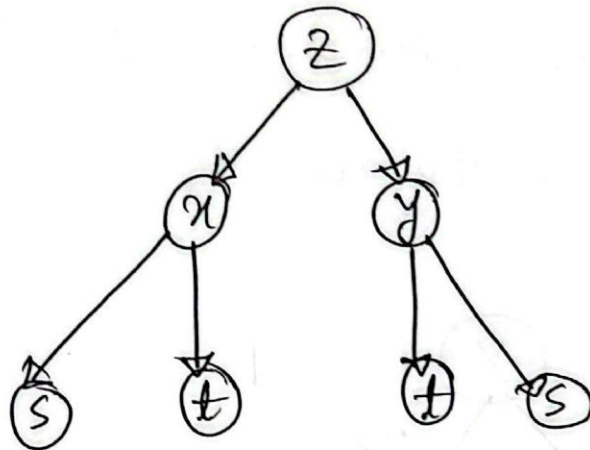
$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} ;$$

* Example:

If $z = e^x \sin y$ where $x = s \cdot t^2$

and $y = s^2 \cdot t$. Find $\frac{\partial z}{\partial s}$, $\frac{\partial z}{\partial t}$.

Ans:



$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$= e^x \sin y \cdot t^2 + e^x \cos y \cdot 2st$$

$$= e^{st^2} \sin(st^2) \cdot t^2 + e^{st^2} \cos(st^2) \cdot 2st$$

$$\frac{\partial z}{\partial s} = t^2 e^{st^2} \sin(st^2) + 2st e^{st^2} \cos(st^2)$$

Ans.

$$\text{and, } \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= e^x \sin y \cdot 2st + e^x \cos y \cdot s^2$$

$$= e^{st^2} \sin(st^2) \cdot 2st + e^{st^2} \cos(st^2) s^2$$

$$\therefore \frac{\partial z}{\partial t} = 2st e^{st^2} \sin(st) + s^2 e^{st^2} \cos(st)$$

Ans.

★ To find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$, we fixed the product of partial derivatives along each path from z to s or z to t , and then add these products.

★

A / f
A A

**** Maxima and Minima of functions
of two variables: (H.A - sec - 13.8)**

*** Definition - 13.8.1:**

A function f of two variables is said to ~~be~~ have relative ~~minimum~~ ^{maximum} at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) that lie inside the disk.

And, f is said to have absolute maximum ~~if~~ at (x_0, y_0) if

$f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f .

* Definition (13.8.2)

A function f of two variables is said to have a relative minimum at (x_0, y_0) if there is a disk D , centered at (x_0, y_0) such that

$$f(x_0, y_0) \leq f(x, y), \quad \forall (x, y) \in D.$$

And f is said to have absolute minimum at (x_0, y_0) if

$$f(x_0, y_0) \leq f(x, y); \text{ for any } (x, y) \in \text{Domain}$$

* Theorem - (13.8.4)

If f has a relative extremum at a point (x_0, y_0) and if the first order partial derivatives of f exists at this point, then

$$f_x(x, y) = 0 \text{ and } f_y(x, y) = 0.$$

* critical point: (13.8.5)

A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a critical point of the function if

$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0$$

or, if one or both partial derivatives do not exist at (x_0, y_0) .

* Saddle point:

A saddle point of a function is a point in the domain of function where it neither attains a maximum value nor attains a minimum value.

* Example:

Find all the critical points of

$$f(x, y) = \frac{x^2}{2} - x + xy^2.$$

Ans:

Given that

$$f(x, y) = \frac{x^2}{2} - x + xy^2$$

$$f_x(x, y) = x - 1 + y^2$$

$$f_y(x, y) = 2xy$$

For critical point

$$f_x(x, y) = 0 \quad \text{and} \quad f_y(x, y) = 0$$

$$\Rightarrow x - 1 + y^2 = 0 \quad \text{--- (1)}$$

$$\Rightarrow 2xy = 0$$

$$\Rightarrow xy = 0 \quad \text{--- (2)}$$

either $x = 0$ or $y = 0$

(Case-1) when $x = 0$

$$\text{From (1)} \quad 0 - 1 + y^2 = 0 \Rightarrow y^2 = 1$$

$$\Rightarrow y = \pm 1$$

(Case-2) when $y = 0$

$$(0, 1); (0, -1)$$

From (1)

$$x - 1 + 0 = 0 \Rightarrow x = 1$$

$$(1, 0)$$

Therefore the critical points
are $(0,1)$, $(0,-1)$, $(1,0)$

★ Theorem - 13.8.6 (The second Partial test)

let f be a function of two variables
with continuous second-order partial
derivatives in some disk centered
at a critical point (x_0, y_0) , and

$$\text{let } D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then
 f has a relative minimum
at (x_0, y_0) .

b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then
 f has a relative maximum at
 (x_0, y_0) .

c) If $D < 0$, then f has a saddle
point at (x_0, y_0) .

d) If $D=0$, then no conclusion can be drawn.

*Example: locate all relative extrema and saddle points of
 $f(x,y) = 4xy - x^4 - y^4$.

Ans:

$$\text{Here } f_x = 4y - 4x^3$$

$$f_y = 4x - 4y^3$$

To find out the critical points, we set,

$$\begin{array}{l|l} f_x = 0 & \text{and } f_y = 0 \\ \Rightarrow 4y - 4x^3 = 0 & \Rightarrow 4x - 4y^3 = 0 \\ \Rightarrow y = x^3 \rightarrow \textcircled{1} & \Rightarrow x = y^3 \rightarrow \textcircled{2} \end{array}$$

$$\begin{array}{l} \text{From } \textcircled{1} \& \textcircled{2} \\ x = y^3 = (x^3)^3 = x^9 \\ \Rightarrow x^9 - x = 0 \end{array}$$

$$\Rightarrow x(x^8 - 1) = 0$$

$$\therefore x = 0 \text{ or } x^8 - 1 = 0$$

$$\Rightarrow x^8 = 1$$

$$\Rightarrow (x^2)^4 = 1 = 1^4$$

$$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\text{When } x = 0, \quad y = x^3 = 0^3 = 0$$

critical point $e_1 \equiv (0, 0)$

$$\text{When } x = 1, \quad y = 1^3 = 1$$

$$e_2 \equiv (1, 1)$$

$$\text{When } x = -1, \quad y = (-1)^3 = -1$$

$$\therefore e_3 \equiv (-1, -1)$$

the critical points for the given functions are

$$(0, 0), (1, 1) \text{ \& } (-1, -1)$$

Here, $f_{xx} = -12x^2$

$f_{yy} = -12y^2$

$f_{xy} = 4$

$\therefore D = f_{xx} f_{yy} - f_{xy}^2 = (-12x^2)(-12y^2) - 4^2$

$\therefore D = 144x^2y^2 - 16$

Number	critical point	D	f_{xx}	conclusion
1	(0,0)	$-16 < 0$		saddle point at (0,0)
2	(1,1)	$128 > 0$	$-12 < 0$	Relative maximum
3	(-1,-1)	$128 > 0$	$-12 < 0$	Relative maximum.

therefore, (0,0) is saddle point and there exists relative maximum at points (1,1) and (-1,-1).

Now, we need to calculate $f(1,1)$ and $f(-1,-1)$.